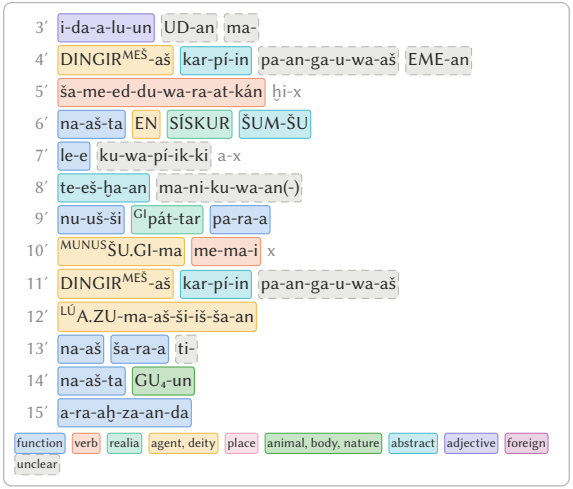
KBo 41.22: an orphan (CTH 470), 33 legible words

Luwian rituals
(Puriyanni, Kuwattalla)
10 reference manuscripts
$s_1 = \cos(\mathbf{x}, \mathbf{c}_1) = \mathbf{0.39}$
nearest tradition

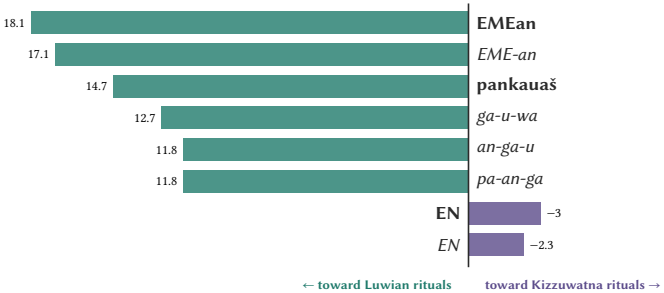
Kizzuwatna rituals
165 reference manuscripts
$s_2 = \cos(\mathbf{x}, \mathbf{c}_2) = \mathbf{0.25}$
runner-up

How a fragment becomes numbers. Every word form and every sign sequence in a word is a feature f, weighted $x_f \propto (1 + \log tf_f)\, idf_f$: frequent here, rare in the corpus. **pankauaš**: idf 6.8; the sign *na*: 1.2. Each tradition is the average $\mathbf{c}_t$ of its manuscripts.

Check by hand. Of the five single manuscripts nearest to it, 5 are Tunnawiya's ritual (CTH 409), which no reference group includes: a lead for the philologist.



Which features pull where: $x_f(c_{1,f} - c_{2,f}) \times 10^3$



All pulls add up to the lead of the nearest tradition:
$m = \sum_f x_f(c_{1,f} - c_{2,f}) = s_1 - s_2 = \mathbf{0.14}$
$m \geq \tau = 0.018 \Rightarrow$ placed with **Luwian rituals**
(τ: the lead at which 80% of test fragments are placed correctly)